

Empirical evidence and state of practice of software agile teams

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ABSTRACT

The paper provides an in depth analysis of empirical evidence on the state of practice within the agile domain obtained through a survey conducted in 2012. The context of focus is agile software processes and teams and the particular topics of interest revolve around three axes: (i) communication; (ii) project management; and (iii) quality assurance and validation. The aim of the survey is to deliver the current levels of agile adoption and practices as these are recorded in the responses of professionals in IT services and the software industry. The goal of the survey is to provide evidence-based assessment of the level of agile adoption by software development organizations, in relation to the general profile of the respondents (country of origin, business sectors, roles, etc.) and compared with different types of practices followed, such as agile techniques adopted, team organization and communication techniques, and project management. Particular patterns and trends are identified in the survey connecting the use of the agile paradigm with the aforementioned practices and investigating its relation with the roles of the respondents and the business strategies of their organizations. Copyright © 2014 John Wiley & Sons, Ltd.

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KEY WORDS: software development; agile methods; teams; adoption; survey; state of practice

1. INTRODUCTION

Agile software development methods [1, 2] have gained an increasing interest by software engineers and researchers worldwide over the last few years. Agile development methods follow a paradigm that advocates collaborative development, with people relations being more important over processes that used to constrain them; lean approaches, with minimizing unnecessary work as much as possible, active customer participation during the development, and, finally, acceptance of change, with uncertainty being part of the development process [3]. However, scientific research reporting empirical evidence based on information gathered through structured surveys is yet scarce in this domain. This paper reports results from an online survey conducted in 2012, which targeted at discovering, analyzing, and presenting empirical findings related to the adoption of the agile approach within software organizations worldwide and the practical implications of this development paradigm on organizational and methodological aspects of software production.

The overarching aim of this work is to help define the current level of agile practice in the software industry, in relation to the demographic features of the participants in the survey, and juxtapose it with business strategies, levels of expertise, and benefits gained by organizations adopting agile methods in

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different countries. Additionally, the primary causes of agile project failures are identified through the testimonies of the respondents—stakeholders and practitioners.

The motivation and significance of this work is supported by the fact that very few surveys have focused on the identification of the critical factors that affect agile adoption. Most of the research reported in literature, thus far, compares traditional and agile approaches and identifies advantages and weaknesses on the use of the agile paradigm through the review of empirical studies [4, 5]. Other researchers, through case studies, have previously identified the need for empirical research frameworks in agile methods in order to make the study of agile development methodologies more comparable [6]. Along these lines, this work investigates the current practice of the agile development approach in three main axes, or sub-processes, namely (i) communication, (ii) project management, and (iii) quality assurance and validation. It also attempts to identify the main practical benefits gained by adopting the agile development approach, as well as to define the main causes of failures occurring in agile projects. The most significant result of our study is the fact that empirical testimony states that the adoption of agile methods significantly enhances productivity and efficiency of the development process, while it also improves the levels of software and process quality.

The rest of this paper is structured as follows: Section 2 presents a short literature overview on empirical studies reporting evidences on the agile paradigm adoption and use. Section 3 describes the methodology followed to collect the data samples, along with a general demographical analysis of the participants and outline of the structure of the collected information. Section 4 presents the empirical findings, focusing on some sections of the questionnaire used, such as the geographical distribution and dispersion of teams, business strategies, levels of expertise, and reflections on the adoption of agile methods. A short evaluation of the results as regards statistical significance and reliability is provided in Section 5, while some threats to their validity are reported in Section 6. Finally, Section 7 summarizes the paper and describes the main conclusions and future research steps.

2. LITERATURE OVERVIEW

The related literature on agile methods adoption gives birth to two main contestants with regard to what is the trend. On one hand, it is the claim that the agile paradigm is here to stay and is a mainstream in the industry and, on the other hand, that it is a declining trend [7]. In previous work of Chan and Thong [8], it was mentioned that agile methodologies adoption is typically resisted by the IT personnel. Industry surveys [9] have also indicated that the acceptance levels of agile methodologies within organizations is at a very small degree, and there is still ample room for improvement. The main obstacles reported in a previous study about agile methodologies adoption were related to difficulties in changing the way of working, the upper management's opposition to carrying out this change, and significant lack of documentation, familiarity, and expertise with the methods [10].

Most of the research volume is concerned with the early stages of agile adoption and use, while small evidence exists on its effective use beyond the initial adoption stage [11]. The benefits related to agile methods are also not identified explicitly yet, and factors mainly related to the resulting high quality in the products developed, increased productivity, transparency, and collaboration effects, are recently reported as anecdotal [7].

In their recent review study, Dybå and Dingsoyr reported that even though the majority of publications (76%) focus on extreme programming, they also mention an urgent need of the studies to include other agile methods in their analyses [5]. The dependency and influence of agile methods on other qualities of the software product and process is fairly analyzed until today.

Regarding agile teams and projects, both of these topics were listed in the top 10 significant research questions obtained from practitioners, as a result of a panel discussion [12]. Among other topics, team effectiveness, costing, and measures, were identified as high priority topics, together with agile and large projects, barriers to self-organization, team location and distribution, trust, success rates, release length, cost expenditures, individuals' personalities, distributed agile, and the role of architecture. The issues related to agile software development are the ones practitioners would like to be researched the most.

Several case studies have reported the main challenges in the adoption of agile methods, such as ignorance of what ‘agility’ means in reality [13] and what kind of broader implications in organizational change it causes [7]. Especially, in complex systems development and in large-scale enterprises, a holistic approach is required [14]. However, in order to understand the requirements of the holistic approach, industrial evidence is required to be collected and reported.

Moreover, evaluating and criticizing the success of this holistic agile approach in enterprises is mentioned in a very small research scale. Therefore, the motivation of the present study to provide an in depth analysis of empirical evidence on the state of practice within the agile domain with respect to IT professionals’ beliefs on the axes of communication, project management, and quality assurance and validation is well justified, especially when analyzed from the viewpoint of particular roles, geographical locations, and contexts.

3. DESIGN OF STUDY

The instrument used for measurement [15] in this study was a specially prepared questionnaire, which was structured to include the following five sections: (A) personal information, roles, and experience; (B) company/organization information; (C) team members and communication; (D) knowledge and strategies; and (E) agile adoption. The questionnaire included 27 questions in total; section A consisted of two questions, section B of three, section C of four, section D of six, and section E of 12, the latter denoting the focus of the survey.

3.1. Survey demographics

The study was conducted between February and March 2012 by means of an online questionnaire. The potential respondents were invited through direct email invitations and postings in mailing lists related to agile development. Many of them were contacted through the Agile Development Manifesto webpage (<http://agilemanifesto.org/>), which provides an open list of supporters. A total of 5900 people were contacted, and 377 responses (6.38%) were obtained. These responses were then filtered so as to include only those respondents that declared to use agile methods. The general profile of the respondents is presented in Section 4, and the assessment of the validity and reliability of their responses is analytically described in Section 5.

3.2. Survey objectives

The analysis of the results obtained from the survey aims at providing empirical evidence on specific aspects of the current agile practice organized in the following categories: (i) general profile—demographics; (ii) organizational and business aspects; and (iii) operational and methodological aspects. To this end, this paper will focus only on those parts of the questionnaire that are relative to the aforementioned categories. The context of the questions included in each category is briefly described as follows:

3.2.1. General profile—demographics

1. What is the profile of the respondents in terms of role and experience?
2. What is the geographical location and distribution scheme of teams in agile organizations?
3. Which agile methodologies and techniques are typically followed?
4. How many agile projects have been developed by the companies participating in the survey?
5. What type of general goals do agile teams usually set?

3.2.2. Organizational and business aspects

1. What is the typical number of developers in agile organizations?
2. Which business sectors are most frequently targeted by developers following agile practices?
3. What is the level of knowledge and expertise on agile methods met in organizations?
4. What are the major causes for unsuccessful agile projects?
5. What are the success drivers of agile methods’ adoption?

3.2.3. Operational and methodological aspects

1. Are there any software quality standards typically used by agile organizations?
2. Which communication techniques are usually followed by agile teams?
3. What is the frequency with which agile teams communicate?
4. What is the typical delivery frequency to customers of agile teams?
5. What type of testing strategies do agile teams usually adopt?
6. What progress control strategies do agile teams usually follow?

The next section presents the empirical evidence gathered for each question and highlights the main findings for each category.

4. SURVEY ANALYSIS

4.1. General profile—demographics

Table I depicts the roles of the participants in the survey. The figures show that the majority of the respondents were project managers ($N=140$, $f=55\%$), team leaders ($N=129$, $f=50\%$), developers ($N=112$, $f=44\%$), designers/architects ($N=107$, $f=42\%$), development managers ($N=95$, $f=37\%$), and programmers ($N=93$, $f=36\%$).

As regards experience, the majority of the respondents had above 10 years (almost 75%) and another 30% 6–10 years. Therefore, the responses to key questions, such as the major causes for successful/unsuccessful agile projects or the success drivers of agile adoption that follow, can be considered truthful and trustworthy as they are sourced from highly experienced practitioners.

The prevailing geographical locations of the participating organizations appear in Figure 1. The USA seem to lead with more than 20% of the respondents originating from there, followed by Europe, where the UK and Germany are the leading countries. India follows UK in close proximity while Cyprus, even though small as a country, appears in the figures because of a number of personal acquaintances and targeted communications that the authors had with agile organizations.

Regarding team distribution, the majority of team members in agile teams are working in locations with more than 3 hours time zone difference away ($N=87$, $f=34\%$). It is interesting to note that another significant percentage (nearly 46%) is co-located (i.e., in the same workspace ($N=63$, $f=25\%$) or building ($N=29$, $f=11\%$) or within a short driving distance ($N=26$, $f=10\%$)). It becomes evident

Table I. Roles of the respondents (multiple roles were also allowed as answer).

Role	Reponses
Project manager	140 (55%)
Team leader	129 (50%)
Developer	112 (44%)
Designer/architect	107 (42%)
Development manager	95 (37%)
Programmer	93 (36%)
Consultant/trainer	66 (26%)
IT manager	51 (20%)
Business analyst	50 (20%)
QA/tester	35 (14%)
Database expert/administrator	33 (13%)
Technical support	32 (13%)
Operations	18 (7%)
IT staff	17 (7%)
GUI designer	15 (6%)
Multimedia	5 (2%)

GUI, graphical user interface.

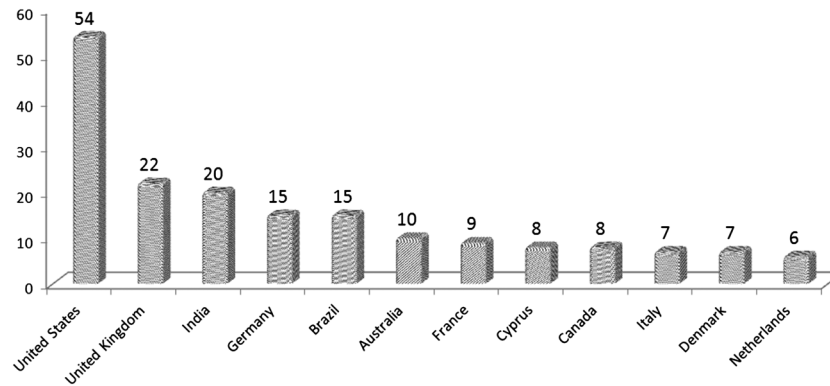


Figure 1. Geographical distribution of responders.

that agile teams do not follow a standard pattern as regards location; they work equally well under a distant or co-located scheme.

The agile methodologies that are typically followed by the companies of the respondents are shown in Figure 2. It is clear that Scrum is the favorite methodology, followed by extreme programming (third most favorite) and hybrid forms of the two (fourth top). It is also worth noting that the second most widely used methodology is a customized approach, which is developed in-house to meet the working environment and characteristics of the companies ('Home-Grown').

With respect to the use of agile approaches, the answers to the question 'Which agile techniques are typically followed?' did not simply promote one or two techniques but suggested that quite a few of them receive preference, a fact that underlines the diversity in the current practices followed (Table II).

The majority of the companies that participated in the survey (almost 50%) have developed a relatively small number of agile projects (between 3 and 10), while only 26% have undertaken a larger number of projects (above 20). Therefore, on the one hand, this finding suggests that at present, the adoption of agile as the main development trend in the organizations is still at its infancy and, on the other hand, that the majority of the reported empirical evidences we are reporting is relying on experiences from a relatively small number of projects executed. Nevertheless, as the type of those projects was not among the key factors of this study, we cannot safely argue about project size and complexity, which affect the experience and expertise gained along the development activities.

The final question that completed the general profile of the respondents was related to the type of general goals that agile teams usually have. The answers summarized in Table III reveal that the

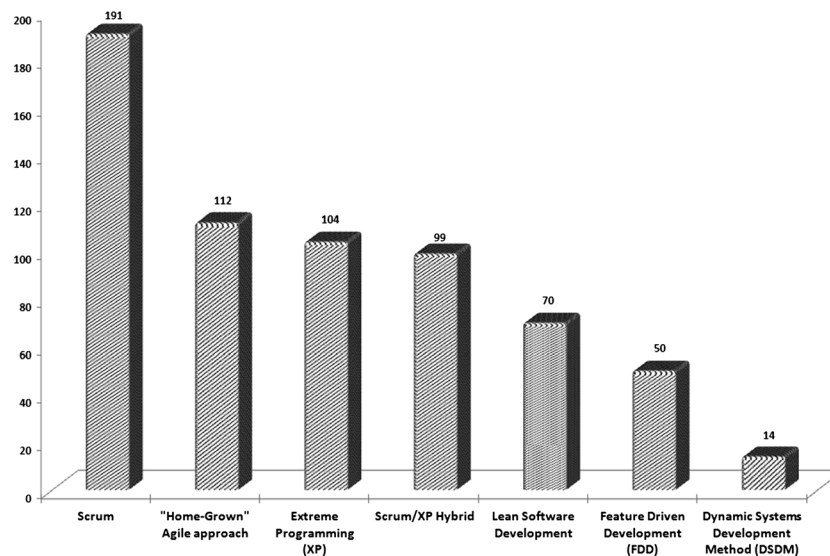


Figure 2. The frequency of use for various agile methodologies.

Table II. Agile techniques used (multiple answers were also allowed).

Agile technique	Responses
Continuous integration	187 (73%)
Iteration planning	180 (71%)
Daily stand-up	177 (69%)
Unit testing	158 (62%)
Refactoring	151 (59%)
Automated builds	149 (58%)
Test-driven development	143 (56%)
Burndown charts	141 (55%)
Release planning	128 (50%)
Analog and digital taskboards	119 (47%)
Collective code ownership	118 (46%)
Pair programming	118 (46%)
Open work area	103 (40%)
Kanban	85 (33%)
Velocity charts	84 (33%)
On-site customer	80 (31%)
Behavior driven development	54 (21%)

Table III. General goals of agile teams (multiple answers were also allowed).

General goal	Response
Produce working software in every iteration/sprint/phase	224 (88%)
Immediate feedback from customer	131 (51%)
Do the simplest thing that could possibly work	116 (45%)
Discuss in groups the goals of the project	106 (41%)
Deliver a fully functional software	99 (39%)
Deliver software keeping a strict schedule	74 (29%)

majority of the agile teams target mostly at producing functional software within the incremental iteration/sprint/phase, followed by receiving immediate feedback from the customer. Also, quite important goals targeted by agile teams are to offer functional simplicity and make the software work with minimum effort, and to share views of the goals through group discussions.

4.2. Organizational and business aspects

The typical number of developers recorded in agile teams is small (between 1 and 10) in one-third of the participating organizations, while we also observe a percentage a little above 10% of quite large development units (above 500 persons). Therefore, we may infer that the agile paradigm is adopted by small teams, but also by large ones. Developers following agile practices are most frequently involved with the IT consulting/services business sector, both technology and software. Other business sectors that follow with comparable frequency include e-commerce systems, banking, government, business services, communications, financial/property, education, medical, and health care.

A significant part of the survey aimed at revealing the level of knowledge and expertise on agile methods in software organizations that are practicing this development approach on a daily basis. The evidence collected showed that above 70% of the practitioners have very good (extensive) knowledge on agile methodologies, while only a very limited number of organizations (2%) report only limited knowledge; 34% of the respondents have 3–4 years practicing agile methods, another 31% has 5–10 years and only 10% above 10 years. Combining the aforementioned measures of expertise with the overall experience reported in the general profile, the majority of the sample has rich experience and expertise, a fact that enhances the significance of the responses as these reflect a mature part of the software industry. This is particularly important as the following two questions, which are quite subjective, will be approached from the perspective of experts or advanced

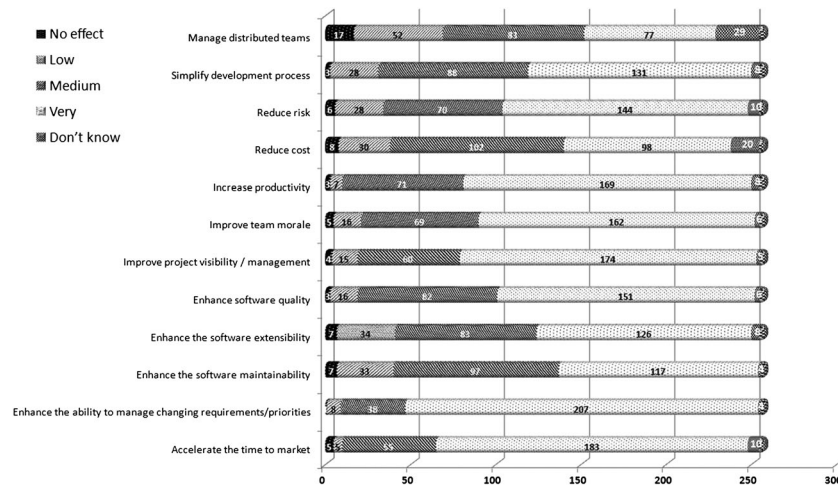


Figure 3. Benefits gained from implementing agile methods.

practitioners of the agile paradigm: (i) What are the success drivers of agile methods' adoption? and (ii) What are the major causes for unsuccessful agile practice? Figure 3 sheds light to the first question by listing the primary benefits from adopting agile methods in the respondents' organizations. Most of the agile benefits received high positive scores (from medium to very high effect) indicating that the respondents have experienced a positive impact from the use of the agile paradigm in their daily development operations and ways of working. The benefits of accelerating time to market and increased ability to manage changing requirements or priorities seem to be the most significant among all.

Table IV depicts the most frequent answers to the second question. It is evident that knowledge and expertise, as well as proper training, are among the top reasons of failure, but nontechnical and managerial aspects play also a crucial role.

4.3. Operational and methodological aspects

This part deals with communication and validation issues, as well as some project management and quality aspects.

Starting with communication issues, the techniques usually used in agile teams are presented in Table V. Face-to-face conversations and email communications are the most popular means of communication, as it is typically the case with teams following traditional development approaches, and they seem to be used within the teams of nearly every respondent. Quite popular are also short meetings and online or audio conference calls. The communication frequency was also addressed by another relevant question and yielded an impressive 70% of the responses for regular intraday meetings. The frequency of delivery to customers proved the relevant theory with periodic increments of the order of 2–3 weeks at a 41% level, followed by 26% for 1 month and 18% for less than 3 weeks. The general tendency therefore dictates frequent deliveries of up to 1 month.

Table IV. Leading causes of unsuccessful agile projects (multiple answers were also allowed).

Reason for failure	Response
Lack of experience or knowledge with agile methods	126 (49%)
External pressure to follow traditional waterfall methods	124 (49%)
Company philosophy or company culture that odds with core agile values	117 (46%)
Insufficient team training	103 (40%)
Lack of management support	97 (38%)
Unwillingness of team to follow agile methods	72 (28%)

Table V. Communication techniques for agile teams (multiple answers were also allowed).

Technique	Response
Face-to-face conversation	232 (91%)
Email	231 (91%)
Meetings (approximately 15 min)	202 (79%)
Online chat	170 (67%)
Audio conference calls	165 (65%)
Repository of shared artifacts	148 (58%)
Follow-up meetings (more than 15 min)	144 (56%)
Video conference calls	134 (53%)
Physical task board	123 (48%)
Electronic task board	122 (48%)
Project dashboards	112 (44%)

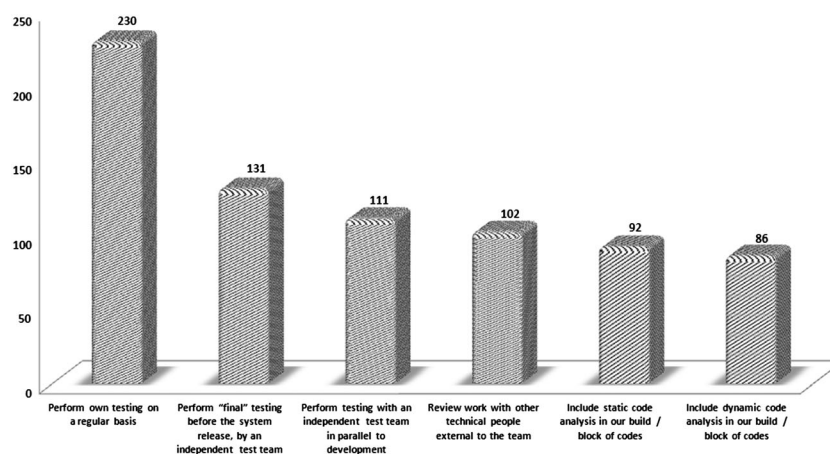


Figure 4. Testing strategies followed by agile teams (multiple answers were allowed).

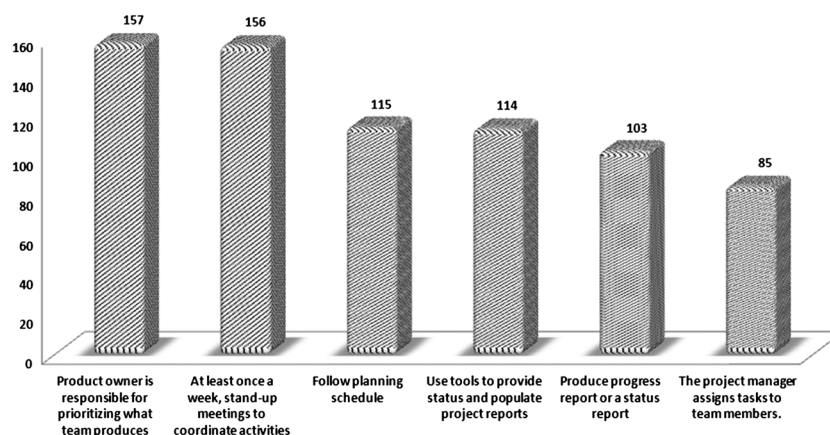


Figure 5. Prioritization and control strategies followed by agile teams (multiple answers were allowed).

When it comes to testing, agile teams usually adopt a variety of strategies, as depicted in Figure 4, which include at a large percentage the strategy of performing their own testing on a regular basis. In addition, at least half of the respondents declared that testing is also performed with the aid of independent teams, either during development or at the final (prior to release) stage.

Another interesting graphical representation of development strategies related to project prioritization and control is provided in Figure 5. More than 60% of the respondents stated that the

customer sets the priorities of the next deliveries and that at least once a week stand-up meetings are held particularly for coordinating the activities of the project. Additionally, nearly half of the organizations participating in the survey follow a schedule for planning their activities. The same percentage reports the use of tools for tracking progress and for populating status reports. A slightly smaller percentage (40%) produces progress or status reports. Finally, the project manager assigns tasks to developers only in one-third of the cases.

The last issue examined was the use of software quality standards (e.g. ISO9000, SPICE, CMM, and CMMI). We notice that only half of the agile teams that participated in the survey declared to follow some quality standards. This observation shows that there is lack of standardization or even standardization experts in agile teams or in the agile context in general, which could be conceived as a challenge but also a threat to the quality assurance of the approach itself.

5. SURVEY EVALUATION

The evaluation of the survey is carried out through measures of sample size, validity, and reliability for multiple answers expressed on a Likert scale [16–18].

Crimp [19] reports the following formula for determining the size n of the sample for random sampling:

$$n = \frac{p \ q Z^2}{e^2} \quad (1)$$

where Z is the abscissa of the normal curve that cuts off an area w at the tails ($1 - w$ equals the desired confidence level, e.g., 95%), e denotes the desired level of precision, p is the estimated proportion of an attribute that is present in the population, and q is $(1 - p)$. The value for Z is found in statistical tables, which contain the area under the normal curve. In our case, the total number of our respondents was 377, 256 of which declared the use of agile methods. Therefore, our p was equal to 0.68 ($q=0.32$), and with $Z=1.96$ and $e=0.05$, we calculated n to be equal to 334 samples. Therefore, our sample size was more than adequate.

As regards reliability, we resort to using coefficient alpha, which was introduced by Lee Cronbach in 1951 [20] so as to provide a measure of the internal consistency of a test or scale expressed as a number between 0 and 1. Internal consistency describes the extent to which all the items in a test measure the same concept or construct, and it therefore assesses interrelatedness of the items within the test. Mathematically, reliability may be defined as the proportion of the variability in the responses to the survey as a result of differences in the answers of the respondents. In other words, answers to a reliable survey will be different because respondents have different opinions and not because the survey is not clear enough or has multiple interpretations.

The computation of Cronbach's alpha is based on the number of items on the survey (N) and the ratio of the average inter-item covariance to the average item variance among the items.

$$\alpha = \frac{N \ \overline{cov}}{\overline{var} \ (N - 1) \ \overline{cov}} \quad (2)$$

Higher values of the alpha coefficient indicate higher reliability for multi-point scales (i.e., rating scale: 1 = poor, 5 = excellent). Nunnally [21] has indicated 0.7 to be an acceptable reliability coefficient, but lower thresholds are also used in the literature, which suggest as a rule of thumb that $\alpha \geq 0.9$ is considered 'excellent', $0.7 \leq \alpha < 0.9$ 'good', $0.6 \leq \alpha < 0.7$ 'acceptable', and $0.5 \leq \alpha < 0.6$ 'poor'.

We used Cronbach's alpha in two ways: First, to assess reliability and consistency of answers on the benefits gained from implementing agile methods, which, on one hand, was the only question of the survey with multiple answers following a specific scale, and on the other hand, it is regarded as one of the most significant questions of the survey. Second, to relate different, yet highly similar in conceptual terms, questions, by joining their answers and treating them as a unified entity.

The first assessment for the benefits of the agile adoption yielded α equal to 0.820, which suggests a fairly good reliability. Further investigation with principal components analysis presented in Table VI revealed that the eigenvalue of the first component, which is related with the acceleration of the time to market, is quite larger compared with those of the rest of the factors and accounts for 35.12% of the variance. Also, the eigenvalues of the second and third components that are related to the ability to manage changing requirements/priorities, and enhancing software maintainability respectively, are also large, the former explaining 12% of the variance and the latter 9.4%.

The second assessment involved combining the answers of three different questions that had similar scope or viewpoint. The combined question cases investigated were (A) the experience with agile methods with the time period of their use, (B) the time between deliveries and sprints, and (C) work experience with the time of agile practice. The corresponding α values are listed in Table VII. We observe that none of the aforementioned three combinations yielded a high alpha coefficient value. Instead, all three were confined to figures below the threshold of 0.6, which is considered poor. This finding suggests that the relations assumed are not so profound in the responses of the survey.

6. LIMITATIONS AND POSSIBLE THREATS TO VALIDITY

The current work presented opinions about agile adoption levels and practices, as these were recorded in the responses of professionals in IT services and the software industry. Some limitations and possible threats to validity of its findings are discussed in this section.

One limitation of the study is associated with the restricted scoping and availability of the information sources. As our study reports empirical evidence, the results obtained inevitably include only the analysis of the respondents' subjective opinions, as it happens with every empirical study. The responses are also impacted by the recent experiences of the participants in agile projects, and thus, we do not know how the results would change if the study was repeated in another period. The responses were also somehow affected by the fact that some of the respondents did not have an

Table VI. Principal components analysis of the agile adoption factor benefits.

Component	Eigenvalues		
	Total	% of variance	Cumulative %
Accelerate the time to market	4.214	35.120	35.120
Enhance the ability to manage changing requirements/priorities	1.440	12.003	47.123
Enhance the software maintainability	1.128	9.396	56.519
Enhance the software extensibility	.903	7.524	64.043
Enhance software quality	.796	6.635	70.678
Improve project visibility/management	.784	6.531	77.209
Improve team morale	.624	5.196	82.405
Increase productivity	.531	4.426	86.831
Reduce cost	.508	4.233	91.065
Reduce risk	.432	3.599	94.664
Simplify development process	.386	3.216	97.880
Manage distributed teams	.254	2.120	100.000

Table VII. Cronbach's alpha values for three question combinations.

Combination	α
A	0.579
B	0.574
C	0.572

extensive project portfolio. To obtain a more representative sample for analysis, it would make sense to repeat the study and conduct a comparison between the different samples. However, this is a research direction to be taken in the future.

In addition, even though our sample size is satisfactory, it may not reflect the industry norm. One may argue that the direct access of the authors to some key respondents (i.e., those having Cyprus as their country of origin) may constitute an additional risk associated with the bias of the data collected. Our rationale for selecting participants that we knew for a fact that they were using the agile paradigm and were considered experts essentially minimized this risk.

Other limitations of this survey relate with the fact that a structured questionnaire was used, and respondents were not able to request further explanations or clarifications regarding the questions. However, because the content and structure of the questionnaire was examined in terms of reliability through statistical measures and was found sufficient, this issue does not pose a significant limitation. The questionnaire was also used in a pre-test phase, in which we collected initial feedback from practitioners, that shaped the final form of the structure and guided the selection of the multiple choices of responses included in this final form. Therefore, the effect of this limitation was practically alleviated. Also, because the results of the survey largely conform to the results of previous literature reviews and findings, along with our informal experiences gained during multiple industrial collaborations and interviews conducted with companies, we may conclude that the results presented in this paper do represent the current trends of the state of practice of agile development methods, theory and practice.

7. CONCLUSIONS

This paper presented evidence-based information from the software and IT industry regarding the state of practice of the agile development approach. Using a specially prepared questionnaire as the main instrument, a total of 377 responses were collected from a large pool of potential participants, with 256 of them replying positively to the question whether they use agile methods for producing software. The latter sample was further analyzed, and the results were organized in three major categories: (i) general profile; (ii) organizational and business aspects; and (iii) operational and methodological aspects.

The most significant findings of the analysis may be summarized as follows:

- i. The respondents were highly experienced, presented a variety of working roles, and were located mostly in the USA and Europe.
- ii. Scrum and other customized (company-specific) agile approaches were the most frequently used in the development methods of the sample participating in the study.
- iii. The benefits gained through agile adoption are multiple, with the most significant being the enhancement of the acceleration of the time to market and the ability to manage and prioritize changing requirements.
- iv. Various reasons were found to be responsible for agile project failures, among which the most significant were the lack of knowledge or experience with agile approaches and the resistance to change from traditional development methods to the agile.

Even though some limitations are mentioned in the study, the content, and more importantly the responses to the questionnaire, were examined in terms of reliability through statistical measures and were found sufficient.

Future work will concentrate on identifying relations between the answers to specific questions, which may reveal other significant associations that have not been reported in this work. For example, it would be interesting to investigate the significance and/or variance in roles when answering a question and present the trend among higher and lower levels of the work hierarchy. Continuing in this direction, future work will present the viewpoints of different roles in relation with specific questions (e.g., how project management issues are experienced by different roles, and how these roles and communication strategies relate with the experience or geographic

location/team distribution). In addition, a comparative study may be carried out considering nonagile and agile organizations, and findings can highlight trends and potential directions for future analyses.

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